

Khusanjon Bobokhojaev

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Computer Science & Physics double major (Math minor, 3.9889 GPA) with research experience in experimental physics and large-scale data analysis, plus self-directed work spanning scientific computing, machine learning, embedded systems, databases, networking, and cybersecurity.

EDUCATION

California Lutheran University — Thousand Oaks, CA

Aug 2023 – Expected May 2027

B.S. Computer Science & B.S. Physics, Minor in Mathematics · **GPA: 3.9889** · Dean's List (all terms)

Physics: Physics for Scientists & Engineers I–II, Classical Mechanics, Quantum Physics, Modern Physics, Statistical Physics & Thermodynamics, Advanced Experimental Physics, Geometric & Physical Optics, Applied Electronics, Physical Modeling with MATLAB, 3D Engineering Design

Computer Science & Data: Intro to Computer Programming, Advanced Programming with Python, Object-Oriented Design & Analysis, Competition Problem Solving, Data Communications & Networks, Database Management Systems, Machine Learning

Mathematics: Calculus I–III, Differential Equations, Discrete Mathematics

RESEARCH EXPERIENCE

Undergraduate Research Fellow (REU) — Washington State University, Pullman, WA

May 2026 – Present

Organic photovoltaics device physics · PI: Prof. Brian Collins

- Investigating charge generation in ternary organic photovoltaic (OPV) solar cells via temperature-dependent time-delayed collection field (TDCF) measurements, targeting charge-generation activation energies at temperatures down to ~80 K.
- Performed low-temperature current–voltage characterization (~173 K) quantifying changes in fill factor, external quantum efficiency, power conversion efficiency, and open-circuit voltage.
- Applying an Arrhenius/Boltzmann framework to extract activation energies and interpret charge-transfer-state dissociation; building analysis and visualization workflows for multi-temperature device datasets.

Research Fellow, CMS Experiment — CERN / Fermilab via California Lutheran University

May 2024 – Dec 2025

Supersymmetry search · Advisor: Dr. Sebastian Carron Montero

- Searched for supersymmetric top-quark (stop) production in dilepton final states ($e\mu/\mu\mu$ channels) using CMS/LHC collision data.
- Built event-selection pipelines and a boosted decision tree (BDT) classifier reaching >99.3% accuracy across three stop-mass hypotheses.
- Completed b-tagging and jet energy correction (JEC) systematic studies; migrated analysis workflows from MiniAOD to NanoAOD, reducing data footprint and improving throughput.
- Developed Python and ROOT (C++) data-processing pipelines and automation on Linux compute environments.

PRESENTATIONS & POSTERS

Poster — “Event Selection & Machine Learning for Detecting SUSY Events in CERN Collision Data.” Southern California Conference for Undergraduate Research, CSU Channel Islands (2025); CLU Undergraduate Research Symposium (2025).

Poster/Talk — “Machine Learning for Detecting SUSY Events in CERN-CMS Collision Data.” CLU Undergraduate Research Symposium (2024).

TECHNICAL PROJECTS

Scientific Computing & Machine Learning

N-Body Gravitation Simulation (C++, Raylib) — real-time gravitational simulator using numerical integration, with interactive visualization and a modular, performance-tuned architecture.

Random Walk Simulation (C++, Python) — statistical simulation with a CSV data pipeline and Python visualization workflow.

Handwritten Digit Recognition (Python, PyTorch) — CNN classifier deployed in an interactive GUI supporting real-time inference and user-generated training data.

Computational Physics in MATLAB (MATLAB) — numerical modeling and simulation projects from Physical Modeling with MATLAB coursework.

Software, Web & Databases

Real-Time Chat Application (Flask, MySQL, Socket.IO) — multi-room chat with user accounts; deployed to PythonAnywhere with environment-based credential management.

CommandoPOS Database System (MySQL, DynamoDB) — point-of-sale database implemented in both relational (MySQL: stored procedures, triggers, views, OLAP star schema) and NoSQL (DynamoDB) backends.

Self-Hosted Cloud Storage Server (Nextcloud, Linux) — deployed and configured a personal file-sync and storage server for remote access.

Terms of Service Analyzer — **Chrome Extension** (JavaScript, Chrome API) — Startup Weekend project (Lead Developer); flags risky ToS clauses via keyword/NLP logic and in-page highlighting.

Personal Portfolio Website (HTML, CSS, JavaScript) — responsive site showcasing research and projects.

Embedded Systems, Robotics & Electronics

SCARA Dual-Linkage Robotic Arm (in progress) — designing a two-link SCARA arm with inverse-kinematics motion planning and stepper/servo control.

IoT Garage Door Controller (ESP32, Blynk) — remote motor control and status monitoring via mobile app, with a custom 3D-printed enclosure.

RC Communication & Control System (Arduino, RF) — low-latency wireless joystick control of motors/servos with battery monitoring and feedback.

Dual-Mode Distance Measurement (Arduino, VL53L0X ToF + ultrasonic) — switchable hybrid sensing for improved accuracy with real-time readout.

OLED Environmental Monitor (Arduino, SSD1306, DHT11) — compact display of time, temperature, humidity, and volume level.

Analog Polyphonic Keyboard (555 timers, LF411, MOSFETs, LTSpice) — 5-key, 3-voice analog synth designed and simulated before hardware build.

Cybersecurity & Networking

WPA2 Handshake Capture & Cracking (Kali Linux, aircrack-ng, Hashcat) — captured a WPA2 handshake in a controlled lab and recovered the passphrase via offline attack.

SQL Injection Lab (DVWA) — performed SQL injection attacks against the Damn Vulnerable Web Application to study input-validation and database security flaws.

Offensive Security Lab “Blue” (TryHackMe, Metasploit, Nmap) — service enumeration, exploitation, and offline NTLM hash cracking.

Network Configuration & Simulation (Cisco Packet Tracer) — designed and configured network topologies for the Data Communications & Networks course.

TEACHING & WORK EXPERIENCE

Instructor → **Lead Instructor** → **Assistant Center Director** — Mathnasium, Thousand Oaks, CA

Jul 2023 – May 2026

- Promoted twice over three years; as Assistant Center Director, led daily operations, staff coordination, and student progress tracking.
- Designed individualized learning plans and taught students in grades 2–12, from arithmetic through calculus.

Physics Department Assistant & Math Center Tutor — California Lutheran University

Sep 2024 – May 2026

- Prepared and maintained undergraduate physics lab setups; assisted students with experimental procedures and concepts.
- Provided one-on-one and group tutoring in mathematics from algebra through calculus and statistics.

Campus Services Crew Member — California Lutheran University

Sep 2025 – May 2026

- Set up and coordinated logistics for academic, athletic, and community events.

LEADERSHIP & ACTIVITIES

- Physics Club**, California Lutheran University — President (2026–27); Vice President (2025–26)
- Computer Science Club**, California Lutheran University — Co-President (2026–27)
- Society of Physics Students (SPS)** — Member
- Competitive Programming** — SoCal ICPC participant; active on Codeforces

TECHNICAL SKILLS

Languages: C++, C, Python, Java, JavaScript, SQL, HTML/CSS, Bash, MATLAB/Octave, Arduino/C

Scientific & Numerical: numerical methods, curve fitting & parameter extraction, statistical data analysis, machine learning, neural networks, data modeling

Frameworks & Libraries: ROOT, NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyTorch, OpenCV, Flask, Socket.IO, Raylib, Pygame

Embedded & Hardware: ESP32, Arduino; sensors (ToF, ultrasonic, DHT11, OLED); I2C/SPI/PWM; analog/digital circuit design; LTspice; soldering & protoboarding; SolidWorks, Onshape, 3D printing

Data & Databases: MySQL (stored procedures, triggers, views), OLAP, DynamoDB; data formats & I/O (NanoAOD/MiniAOD)

Security & Networking: Nmap, Metasploit, Wireshark, tcpdump, aircrack-ng, John the Ripper, Hashcat; web-app security (DVWA / SQL injection); Cisco Packet Tracer; TCP/IP, DNS/DHCP/ARP/ICMP, HTTP/HTTPS, SSH/VPN/NAT; Kali Linux

Tools & Platforms: Git/GitHub, Linux (Ubuntu, Kali), Windows, VS Code, Jupyter, PlatformIO, Visual Studio, self-hosting (Nextcloud)

Spoken Languages: Uzbek & English (native), Russian (intermediate), Mandarin Chinese (beginner; coursework: Mandarin I–II)

HONORS & AWARDS

Rossing Physics Scholar (\$5,000, 2025) · ALLIES Foundation Research Scholarship (\$4,000, 2025) · VITAL Research Grant (\$4,000, 2024) · Sigma Pi Sigma Physics Honor Society · Dean's List (all semesters)